

Estrutura de arquivos: CSV, Excel e JSON



Plataforma completa de aprendizado
contínuo em programação.

#BoostingPeople

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O que é um arquivo de dados?

É um arquivo que armazena informações organizadas de forma digital. Esses formatos são usados para troca, armazenamento e análise de dados.

Exemplos:

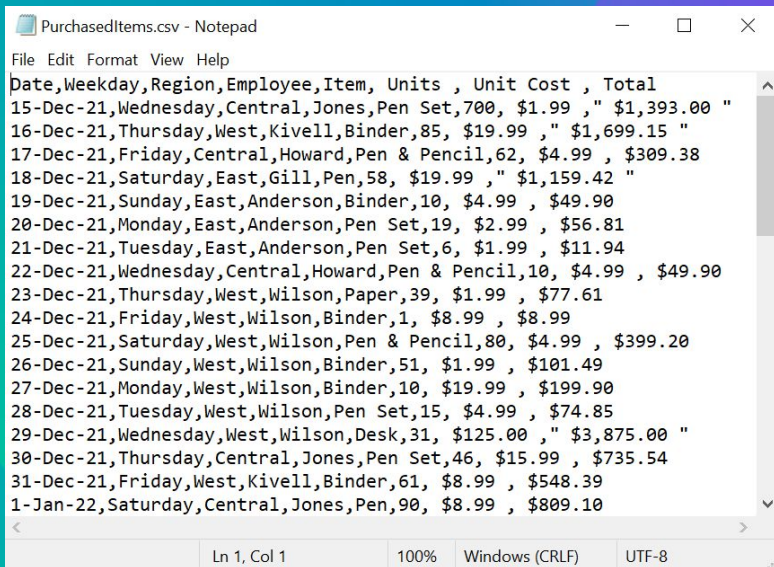
- Tabela de clientes em Excel
- Lista de compras em CSV
- Dados de API em JSON

O que é um CSV?

CSV = Comma-Separated Values (valores separados por vírgula)

- Arquivo de texto simples
- Cada linha representa um registro
- Colunas separadas por vírgula, ponto e vírgula ou outro separador

O que é um CSV?

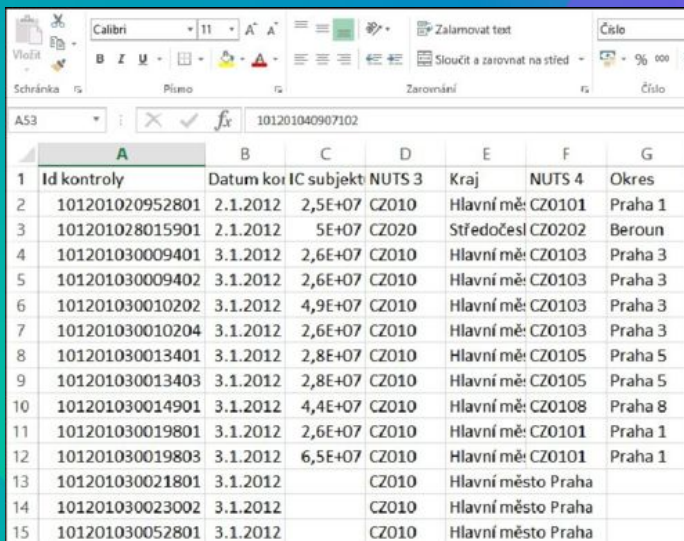


```
PurchasedItems.csv - Notepad
File Edit Format View Help
Date,Weekday,Region,Employee,Item, Units , Unit Cost , Total
15-Dec-21,Wednesday,Central,Jones,Pen Set,700, $1.99 , " $1,393.00 "
16-Dec-21,Thursday,West,Kivell,Binder,85, $19.99 , " $1,699.15 "
17-Dec-21,Friday,Central,Howard,Pen & Pencil,62, $4.99 , $309.38
18-Dec-21,Saturday,East,Gill,Pen,58, $19.99 , " $1,159.42 "
19-Dec-21,Sunday,East,Anderson,Binder,10, $4.99 , $49.90
20-Dec-21,Monday,East,Anderson,Pen Set,19, $2.99 , $56.81
21-Dec-21,Tuesday,East,Anderson,Pen Set,6, $1.99 , $11.94
22-Dec-21,Wednesday,Central,Howard,Pen & Pencil,10, $4.99 , $49.90
23-Dec-21,Thursday,West,Wilson,Paper,39, $1.99 , $77.61
24-Dec-21,Friday,West,Wilson,Binder,1, $8.99 , $8.99
25-Dec-21,Saturday,West,Wilson,Pen & Pencil,80, $4.99 , $399.20
26-Dec-21,Sunday,West,Wilson,Binder,51, $1.99 , $101.49
27-Dec-21,Monday,West,Wilson,Binder,10, $19.99 , $199.90
28-Dec-21,Tuesday,West,Wilson,Pen Set,15, $4.99 , $74.85
29-Dec-21,Wednesday,West,Wilson,Desk,31, $125.00 , " $3,875.00 "
30-Dec-21,Thursday,Central,Jones,Pen Set,46, $15.99 , $735.54
31-Dec-21,Friday,West,Kivell,Binder,61, $8.99 , $548.39
1-Jan-22,Saturday,Central,Jones,Pen,90, $8.99 , $809.10
Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

O que é um XLSX (Excel)?

- Formato nativo de planilhas do Excel
- Suporta múltiplas abas, fórmulas, estilos
- Mais pesado e complexo que CSV
- Não é ideal para automações, mas ótimo para visualização manual

O que é um XLSX (Excel)?



	A	B	C	D	E	F	G
1	Id kontroly	Datum kontroly	IC subjekt	NUTS 3	Kraj	NUTS 4	Okres
2	101201020952801	2.1.2012	2,5E+07	CZ010	Hlavní město	CZ0101	Praha 1
3	101201028015901	2.1.2012	5E+07	CZ020	Středočeský	CZ0202	Beroun
4	101201030009401	3.1.2012	2,6E+07	CZ010	Hlavní město	CZ0103	Praha 3
5	101201030009402	3.1.2012	2,6E+07	CZ010	Hlavní město	CZ0103	Praha 3
6	101201030010202	3.1.2012	4,9E+07	CZ010	Hlavní město	CZ0103	Praha 3
7	101201030010204	3.1.2012	2,6E+07	CZ010	Hlavní město	CZ0103	Praha 3
8	101201030013401	3.1.2012	2,8E+07	CZ010	Hlavní město	CZ0105	Praha 5
9	101201030013403	3.1.2012	2,8E+07	CZ010	Hlavní město	CZ0105	Praha 5
10	101201030014901	3.1.2012	4,4E+07	CZ010	Hlavní město	CZ0108	Praha 8
11	101201030019801	3.1.2012	2,6E+07	CZ010	Hlavní město	CZ0101	Praha 1
12	101201030019803	3.1.2012	6,5E+07	CZ010	Hlavní město	CZ0101	Praha 1
13	101201030021801	3.1.2012		CZ010	Hlavní město	Praha	
14	101201030023002	3.1.2012		CZ010	Hlavní město	Praha	
15	101201030052801	3.1.2012		CZ010	Hlavní město	Praha	

O que é um JSON?

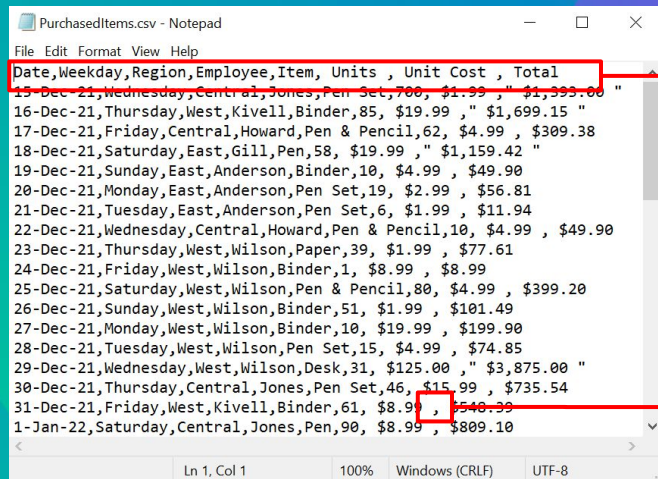
JSON = JavaScript Object Notation

- Formato usado em APIs e dados estruturados em chave-valor
- Pode representar dados mais complexos (listas dentro de listas)

Exemplo:

```
{"nome": "Ana",  
  "idade": 34,  
  "compras": ["calça", "camisa"]}
```

Partes de um CSV



Date	Weekday	Region	Employee	Item	Units	Unit Cost	Total
15-Dec-21	Wednesday	Central	Jones	Pen Set	700	\$1.99	\$1,393.00
16-Dec-21	Thursday	West	Kivell	Binder	85	\$19.99	\$1,699.15
17-Dec-21	Friday	Central	Howard	Pen & Pencil	62	\$4.99	\$309.38
18-Dec-21	Saturday	East	Gill	Pen	58	\$19.99	\$1,159.42
19-Dec-21	Sunday	East	Anderson	Binder	10	\$4.99	\$49.90
20-Dec-21	Monday	East	Anderson	Pen Set	19	\$2.99	\$56.81
21-Dec-21	Tuesday	East	Anderson	Pen Set	6	\$1.99	\$11.94
22-Dec-21	Wednesday	Central	Howard	Pen & Pencil	10	\$4.99	\$49.90
23-Dec-21	Thursday	West	Wilson	Paper	39	\$1.99	\$77.61
24-Dec-21	Friday	West	Wilson	Binder	1	\$8.99	\$8.99
25-Dec-21	Saturday	West	Wilson	Pen & Pencil	80	\$4.99	\$399.20
26-Dec-21	Sunday	West	Wilson	Binder	51	\$1.99	\$101.49
27-Dec-21	Monday	West	Wilson	Binder	10	\$19.99	\$199.90
28-Dec-21	Tuesday	West	Wilson	Pen Set	15	\$4.99	\$74.85
29-Dec-21	Wednesday	West	Wilson	Desk	31	\$125.00	\$3,875.00
30-Dec-21	Thursday	Central	Jones	Pen Set	46	\$15.99	\$735.54
31-Dec-21	Friday	West	Kivell	Binder	61	\$8.99	\$548.39
1-Jan-22	Saturday	Central	Jones	Pen	90	\$8.99	\$809.10

Header: É a linha usada como nome das colunas

Separador: Define como as colunas são separadas no arquivo de texto.